Search in Large Spaces without a Utility Function

Thursday demonstration • UCSC Augmented Design Lab

What are we trying to demonstrate?

Can an LLM generate a new game, express its rules in Ludii, and produce a game that people and AI agents can play?

Generate → Validate → Play → Evaluate

From a game idea to a playable Ludii game

1. Prompt an LLM using example games
2. Generate and revise a .lud description
3. Check that the rules compile and play
4. Load the game and run AI versus AI

Working example: 4×4 Four-in-a-Row

- Two players place X and O on empty squares
- Four in a line wins; a full board can be a draw
- Ludii compiled the game and enforced the rules
- AI versus AI completed; trial saved

The actual Ludii game definition

Four-in-a-Row.lud • 4 × 4 board • four in a line wins

```
(game "Four-in-a-Row"
  (players 2)
  (equipment {
    (board (square 4))
    (piece "Disc" P1)
    (piece "Cross" P2) })
  (rules
    (play (move Add (to (sites Empty))))
    (end {
      (if (is Line 4) (result Mover Win))
      (if (is Full) (result Mover Draw)) })))
```

Game definition tested in Ludii; source is in the GitHub repository.

How to run the game in Ludii

1. Load Four-in-a-Row.lud

2. Select an AI player for each side

3. Restart the game, then press Play/Pause

4. Watch the moves and result; save a .trl trial

Requires Ludii and Java for live play.

An AI-versus-AI trial has already been saved in the repository.

From Tic-Tac-Toe to Qubic

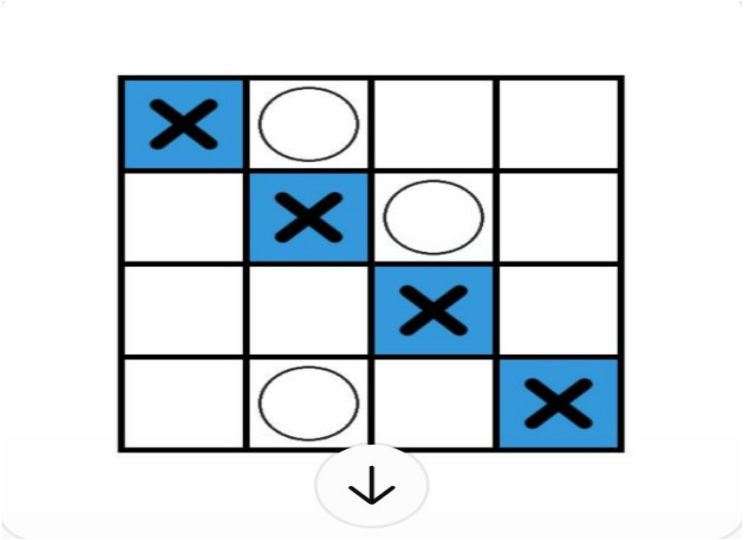
3×3 Tic-Tac-Toe



Draw • 9 cells
$3^9 \approx 2\times10^4$ states

Elementary solution

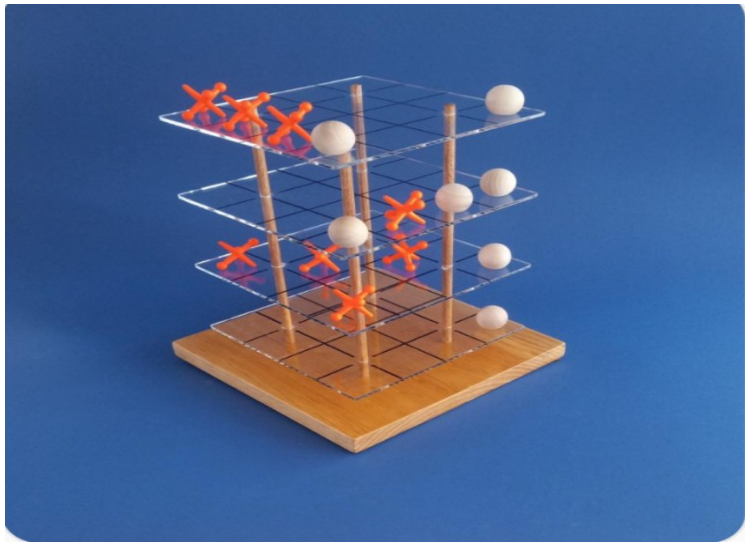
4×4 Four-in-a-Row



Draw • 16 cells
$3^{16} \approx 4\times10^7$ states

Pairing-strategy proof

4×4×4 Qubic



First-player win • 64 cells
$3^{64} \approx 3\times10^{30}$ states

Patashnik (1980): ~1,500 h
Thompson: independent check

What works — and what comes next?

Working now

- Game rules compile
- Legal play and victory verified
- AI versus AI demonstrated
- .lud and .trl files saved

Next steps

- Integrate teammate games
- Automate critique and revision
- Filter invalid games
- Test metrics of strategic interest

Research question: Can game metrics predict enduring interest?